



VULMS - VU Learning Management System



Question # 2 of 2

Time Left

Which of the following would usually require the smallest sample size because of its efficiency?

Select the correct option

[Reload Math E](#)☐

Two stage cluster sampling

☐

Quota sampling

☐

One stage cluster sampling

☐

Simple random sampling

[Click to Save Answer & Move to Next Que](#)



Question # 2 of 2

Time Left

Which of the following would usually require the smallest sample size because of its efficiency?

Select the correct option

[Reload Math E](#)

- | | |
|----------------------------------|----------------------------|
| ☐ | Two stage cluster sampling |
| ☐ | Quota sampling |
| ☐ | One stage cluster sampling |
| ☒ | Simple random sampling |

[Click to Save Answer & Move to Next Que](#)



Time Left: 83 sec(s)

Question # 2 of 2

A representative part of a larger population according to certain rules is called

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|------------|
| ☐ | Statistic |
| ☐ | Sample |
| ☐ | Population |
| ☐ | Element |

[Click to Save Answer & Move to Next Question](#)



Post-Assessment



Question # 2 of 2

Time Left: 79 sec(s)

A representative part of a larger population according to certain rules is called

Select the correct option

[Reload Math Equations](#)

- | | |
|----------------------------------|------------|
| ☐ | Statistic |
| ☒ | Sample |
| ☐ | Population |
| ☐ | Element |

[Click to Save Answer & Move to Next Question](#)

9:55 AM

100% 44



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Post-Assessment

Time Left: 04 sec(s)

Question # 2 of 2

Which of the following sampling methods is the best way to select a group of people for a study if you are interested in making statements about the larger population?

Select the correct option

Reload Math Equations

☐

Convenience sampling

☐

Purposive sampling

☐

Quota sampling

☐

Random sampling

Click to Save Answer & Move to Next Question

9:55 AM

4G 54 80% 100%



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Pre-Assignment

Time Left: 80 sec(s)

Question # 2 of 2

Which of the following sampling methods is the best way to select a group of people for a study if you are interested in making statements about the larger population?

Select the correct option

Reload Math Equations

☐	Convenience sampling
☐	Purposive sampling
☐	Quota sampling
☒	Random sampling

Click to Save Answer & Move to Next Question

**Question # 1 of 2**

Auxiliary variable is also called:

Select the correct option

☐	sampling Variable
☐	dependent variable
☐	None of the above
☐	supporting variable



**Question # 1 of 2**

Auxiliary variable is also called:

Select the correct option

☐	sampling Variable
☐	dependent variable
☐	None of the above
☐	supporting variable



**Question # 1 of 2**

Auxiliary variable is also called:

Select the correct option

☐	sampling Variable
☐	dependent variable
☐	None of the above
☒	supporting variable



**Question # 2 of 2**

Grant estimated the population of England in which year:

Select the correct option

☐	In 1962
☐	In 1667
☒	In 1662
☐	In 1952



Post-Assessment

Question # 1 of 2

Hartley-Ross proposed the unbiased regression estimator in which year:

Select the correct option

☒	In 1954
☐	In 1971
☐	In 1965
☐	None of the above

Click

Post-Assessment

Question # 2 of 2

When Hartley-Ross proposed the unbiased ratio estimator:

Select the correct option

- | | |
|----------------------------------|-------------------|
| ☒ | In 1954 |
| ☐ | In 1964 |
| ☐ | In 1965 |
| ☐ | None of the above |

Click

What is the R command to find out mean of x?



None of the above



mean ($\bar{x}$)



X = mean



Mean ()



Post-Assessment

Question # 2 of 2

What is the R command to find out variance of x?

Select the correct option

☐	· X = variance
☐	· Var ()
☒	· Var (x)
☐	· None of the above

**Question # 1 of 2**

Estimated value of square root of 800 is:

Select the correct option

☐	2.5495
☐	254.95
☐	48.48
☒	28.28



10:02 AM

5G 30% 100%



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Question # 1 of 2

If sample size is large sample mean to approaches to population mean

Select the correct option



True



False

Click to Save Answer





Question # 2 of 2

The estimate of ratio in ratio method of estimation is consistent estimate of population ratio.

Select the correct option

☒	True
☐	False

Click to Save Answer &



Question # 2 of 2

If sample size is large sample mean to approaches to population mean

Select the correct option

☒

True

☐

False

Click to Save Answer





Question # 1 of 2

What is the R command for writing variance-covariance matrix $\Sigma = \begin{bmatrix} 1 & .95 \\ .95 & 1 \end{bmatrix}$?

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | <code>sig=matrix(c(1,1,0.75,0.75),ncol=2)</code> |
| ☐ | <code>sig=matrix(c(1,0.75,0.75,1),ncol=4)</code> |
| ☒ | <code>sig=matrix(c(1,0.75,0.75,1),ncol=2)</code> |
| ☐ | None of the above |

[Click to Save](#)

**Question # 2 of 2**

Double sampling is also known as

Select the correct option

Two stage sampling



Two phase sampling



All of above



Two directional sampling

Click to Save





Post-Assessment

Time Left: 71 sec(s)

Question # 1 of 2

Systematic sampling means

select the correct option

Reload Math Equations

☐

Selection of n contiguous units

☐

Selection of n largest units

☒

Selection of n units situated at equal distance

☐

Selection of n middle units in a sequence

[Click to Save Answer & Move to Next Question](#)

Post-Assessment

Question # 1 of 2

Time Left: 76 sec(s)

Two stage sampling design is better than the single stage sampling only if the elementary units in the same units are

Select the correct option

[Reload Math Equations](#)☒ Negatively correlated☐ Positively correlated☐ Independent☐ Either a or b[Click to Save Answer & Move to Next Question](#)



Pre-Test Assessment



Question # 2 of 2

Time Left: 68 sec

In random sampling, the probability of selecting an item from the population is:

Select the correct option

[Reload Math Equations](#)☐

One

☐

Zero

☒

Known

☐

Unknown

[Click to Save Answer & Move to Next Question](#)

10:18 AM

5G Signal 100%



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Final Assessment

Time Left: 81 sec(s)

Question # 2 of 2

The list of all units in a population is called:

Select the correct option

Reload Math Equations



Sampling frame



Bias



Statistic



Parameter

Click to Save Answer & Move to Next Question

Question # 1 of 2

Time Left: 59 sec(s)

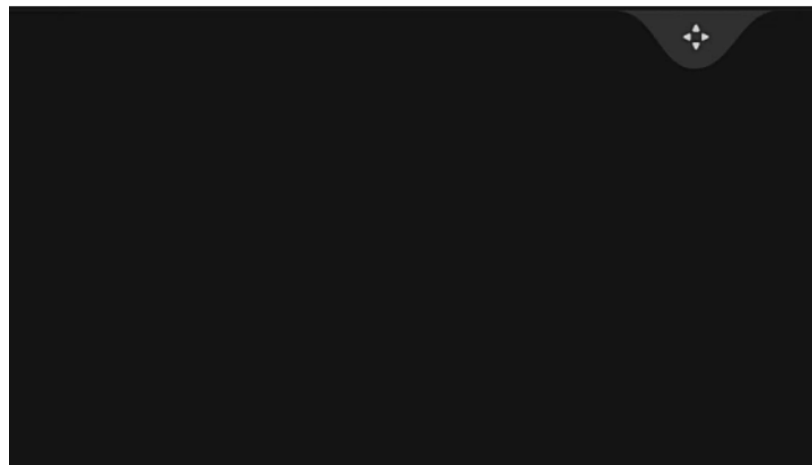
Among these, which sampling is based on equal probability?

Select the correct option

[Reload Math Equations](#)

- | | |
|----------------------------------|----------------------------|
| ☐ | Stratified random sampling |
| ☐ | Systematic sampling |
| ☐ | Probability sampling |
| ☒ | Simple random sampling |

[Click to Save Answer & Move to Next Question](#)



Question # 1 of 2

Time

The distribution that is formed by all possible values of a statistics is known as:

Select the correct option

Reload M

- ☐ Hypergeometric distribution
- ☐ Normal distribution
- ☒ Sampling distribution
- ☐ Binomial distribution

Click to Save Answer & Move to Next

Question # 2 of 2

The difference between a statistic and the parameter is called:

Select the correct option

- ☐ Random
- ☐ Probability
- ☐ Non-random
- ☒ Sampling error

Click to Save Answer & Move



Math Assessment



Question # 2 of 2

Time Left: 7

The difference between the expected value of a statistic and the value of the parameter being estimated is called

Select the correct option

Reload Math Equ



Sampling error



Bias



Standard error



Non-sampling error

[Click to Save Answer & Move to Next Question](#)

Question # 1 of 2

Which of the following is not a type of non-probability sampling?

Select the correct option

- | | |
|----------------------------------|----------------------------|
| ☒ | Stratified random sampling |
| ☐ | Quota sampling |
| ☐ | Convenience sampling |
| ☐ | Snowball sampling |

Click to Save Answer

Question # 2 of 2

Increasing the sample size has the following effect upon the sampling error?

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | It has no effect on the sampling error |
| ☐ | All of the above |
| ☒ | It reduces the sampling error |
| ☐ | It increases the sampling error |

Click to Save Answer

Question # 1 of 2

Sampling error can be reduced by:

Select the correct option

- | | |
|----------------------------------|--------------------------------|
| ☐ | Non-probability sampling |
| ☒ | Increasing the sample size |
| ☐ | Increasing the population size |
| ☐ | Decreasing the sample size |

Click

Question # 2 of 2

Sample is regarded as a subset of?

Select the correct option

☐	Set
☐	Distribution
☐	Data
☒	Population

Click

Question # 2 of 2

Randomized technique was first proposed by S. L. Warner in:

Select the correct option

- | | |
|----------------------------------|------|
| ☒ | 1965 |
| ☐ | 1985 |
| ☐ | 1975 |
| ☐ | 1963 |

Click

Question # 1 of 2

Cluster sampling is used:

Select the correct option

- | | |
|----------------------------------|------------------------------------|
| ☒ | To reduce the cost |
| ☐ | When variance is fixed |
| ☐ | When population units are in group |
| ☐ | To reduce the sample size |

Question # 2 of 2

Q. Cluster sampling is also known as_____ sampling.

Select the correct option

☐	Judgmental
☐	Controlled
☒	Area
☐	Non-probability

C

The list of all units in a population is called:

Select the correct option

☒	Sampling frame
☐	Parameter
☐	Bias
☐	Random sampling

Click to Save Answer & Move to Next Question



Mean of clusters mean is the unbiased estimator of population mean when:

Select the correct option

 Reload Math Equations

☒	Clusters are equal sizes
☐	Both equal and unequal cluster sizes
☐	Clusters are unequal sizes
☐	None of the above

Click to Save Answer & Move to Next Question



Which is the R command for selecting a sample of size 10 from the population `yp <- c(11,150, 121, 198, 12, 136, 14, 129, 17, 115, 186, 110, 121, 15, 14)`:

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|------------------------------------|
| ☐ | <code>S <- sample(yp,1)</code> |
| ☐ | <code>S <- sample(yp,25)</code> |
| ☒ | <code>S <- sample(yp,10)</code> |
| ☐ | <code>S <- sample(yp,5)</code> |

Click to Save Answer & Move to Next Question



Which of the following would generally require the largest sample size?

Select the correct option

 Reload Math Equations

☐	Systematic sampling
☐	Simple random sampling
☒	Cluster sampling
☐	Proportional stratified sampling

Click to Save Answer & Move to Next Question



if
 $Var(\bar{y}_{sy}) < Var(\bar{y}_{srs})$

:

Select the correct option

 Reload Math Equations

☐	$\rho_w = 0$
☐	$\rho_w = \frac{1}{(nk - 1)}$
☐	$\rho_w > -\frac{1}{(nk - 1)}$
☒	$\rho_w < -\frac{1}{(nk - 1)}$

Click to Save Answer & Move to Next Question



Feb 03, 2023

STA632 - Sampling Techniques (Quiz 2)

Question # 5 of 10 (Start time: 02:00:09 PM, 03 February 2023)

What is the bias of ratio estimator in two stage sampling?

Select the correct option

☐ $MSE(\bar{y}_{r2s}) \approx Y^2(V_{02} - V_{20} + 2V_{11})$

☐ $MSE(\bar{y}_{r2s}) \approx Y^2(V_{02} - V_{20} - 2V_{11})$

☒ $MSE(\bar{y}_{r2s}) \approx Y^2(V_{02} + V_{20} - 2V_{11})$

☐ None of the above